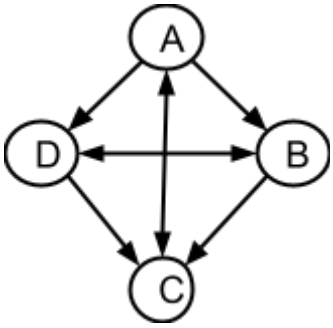
American Computer Science League

2022-2023 ● Contest 4: Short Problems ● Classroom Division

1. Graph Theory

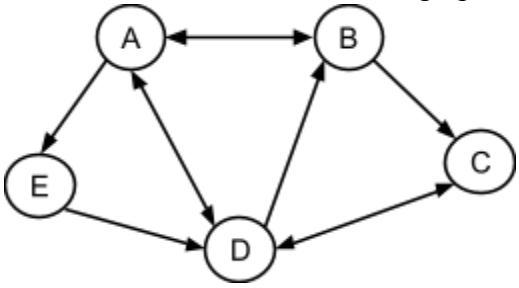
How many paths of length 3 are there in the following directed graph?



A. 20
B. 22
C. 24
D. 26
E. 28

2. Graph Theory

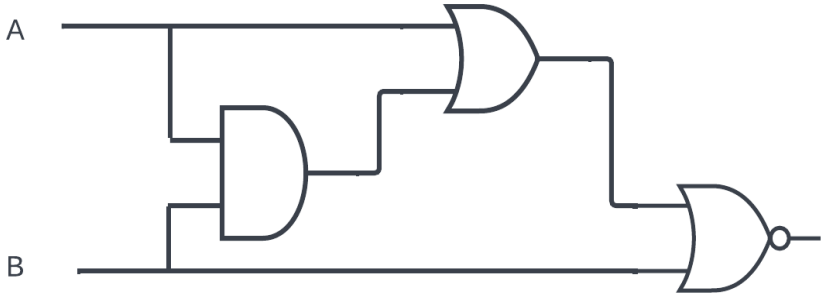
How many cycles are there from A in the directed graph below?



A. 2
B. 3
C. 4
D. 5
E. 6

3. Digital Electronics

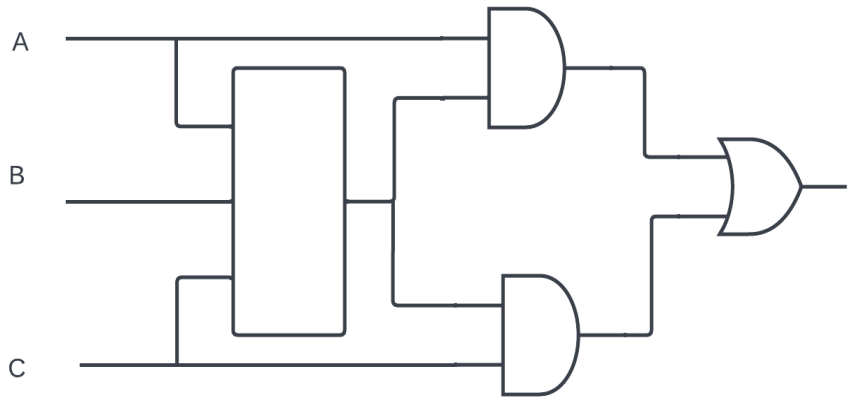
Simplify the Boolean expression represented by the following digital circuit:



A. $\overline{A}$
B. $\overline{B}$
C. $\overline{A} + \overline{B}$
D. $\overline{A}\,\overline{B}$
E. $\overline{A}\,B$

4. Digital Electronics

How many ordered triples make the following digital circuit TRUE if a new gate (rectangle) is defined to be TRUE if exactly 2 of the 3 inputs are TRUE?



A. 0
B. 1
C. 2
D. 3
E. 4

5. Assembly Language

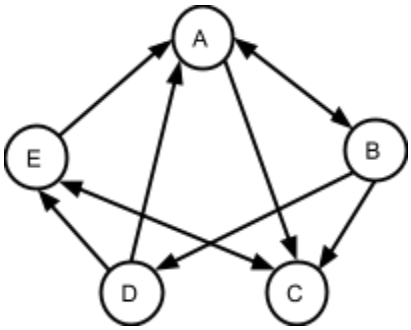
What is the output when the following program is executed?

```
A       DC      0
B       DC      1
TOP     LOAD    B
        MULT    =2
        ADD     A
        STORE   C
        LOAD    =0
        STORE   D
TRY     LOAD    D
        ADD     =1
        STORE   D
        MULT    D
        SUB     C
        BE      STOP
        BL      TRY
        LOAD    B
        STORE   A
        LOAD    C
        STORE   B
        BU      TOP
STOP    PRINT   C
        END
```

A. 29
B. 70
C. 121
D. 169
E. 408

6. Graph Theory

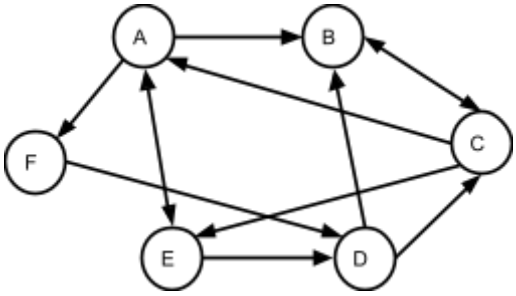
How many paths of length 2 are there in the following directed graph?



A. 15
B. 16
C. 17
D. 18
E. 19

7. Graph Theory

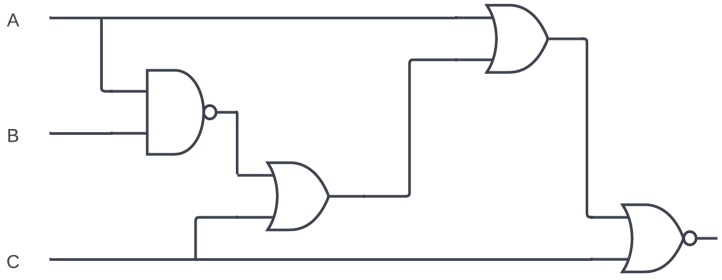
How many cycles are there that do not include A in the following directed graph?



A. 1
B. 3
C. 5
D. 8
E. 11

8. Digital Electronics

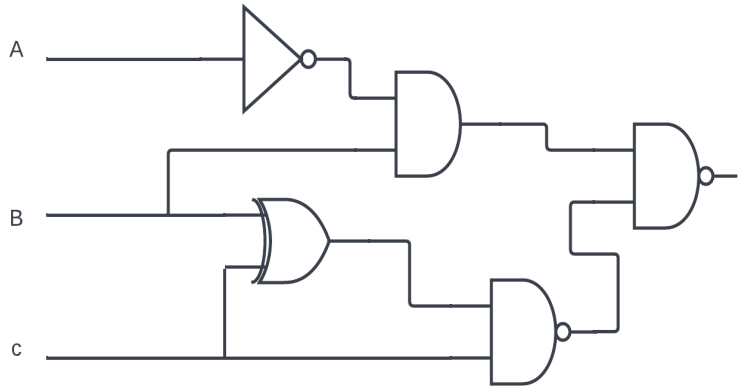
Simplify the Boolean expression represented by the digital diagram below:



A. 0
B. A
C. $\overline{B}$
D. C
E. 1

9. Digital Electronics

How many ordered triples make the following digital circuit FALSE?



A. 0
B. 2
C. 4
D. 6
E. 8

10. Assembly Language

What is the output when the following program is executed?

```
A       DC      1
C       DC      1
        LOAD    C
TOP     MULT    =100
        SUB     A
        BL      STOP
        LOAD    A
        MULT    =2
        STORE   A
        LOAD    C
        ADD     =1
        STORE   C
        BU      TOP
STOP    PRINT   C
        END
```

A. 10
B. 11
C. 12
D. 13
E. 14